Beam analysis using MATLAB

Exercise B

Stiffness method and analytical solution

This guide explains how the MATLAB solution is built and reproduces its result by hand. The direct stiffness calculation and the slope-deflection calculation use the same Euler-Bernoulli beam theory; their agreement checks the assembly, boundary conditions, and signs.

The fixed-end support reaction moments are 333.895 kN m counterclockwise at support 1 and 153.937 kN m clockwise at support 3.

Determine the moments at 1 and 3 if the support 2 settles 30 mm. Assume 2 is a roller and 1 and 3 are fixed. EI=3927 kN*m^2.

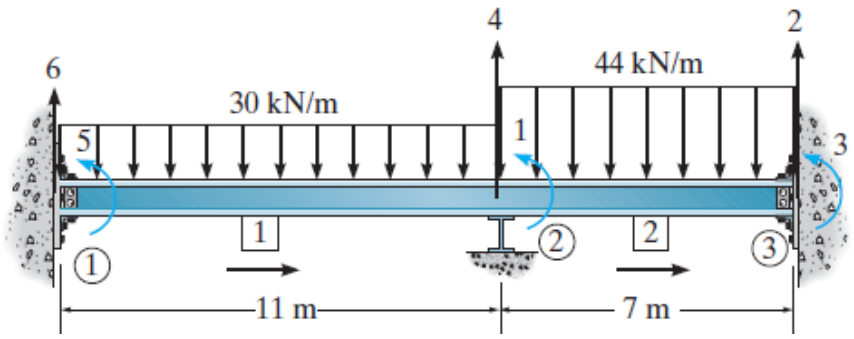


Figure 1. Circled numbers identify physical supports.

Use kN, m, and radians. The model assumes a straight prismatic beam, linear elastic behaviour, small displacements and rotations, and negligible shear deformation. Both ends are fixed; the intermediate roller allows rotation while maintaining continuity of the beam.

1 Define the model and its degrees of freedom

The code creates one beam element per span. Each node has a vertical displacement and a bending rotation. There are six global degree-of-freedom positions, but the boundary conditions leave only the rotation at node 2 unknown.

$$\mathbf{d} = [v_1, \theta_1, v_2, \theta_2, v_3, \theta_3]^T$$

Index	Physical variable	Condition
1	v_1	0
2	θ_1	0
3	v_2	-0.030 m
4	θ_2	unknown
5	v_3	0
6	θ_3	0

MATLAB representation

```
xNode = [0 11 18];
connectivity = [1 2; 2 3];
prescribedDOF = [1 2 3 5 6];
freeDOF = 4;
d = zeros(6,1);
d(3) = -30/1000;
```

xNode stores coordinates, and each row of connectivity stores the node pair of one element. The index rule 2*i-1 selects v at node i; 2*i selects its rotation. Therefore element 1 maps to [1 2 3 4], and element 2 maps to [3 4 5 6]. The common entries 3 and 4 make displacement and rotation continuous at the roller.

```
i = connectivity(e,1);
j = connectivity(e,2);
elementDOF(e,:) = [2*i-1, 2*i, 2*j-1, 2*j];
```

Sign convention: vertical forces and displacements are positive upward; nodal moments and rotations are positive counterclockwise. Internal bending moment is positive in sagging. A downward load is consequently negative. A negative prescribed displacement represents settlement.

A roller cannot supply an external reaction couple. It does not release bending continuity within a continuous beam, so the internal moment at node 2 is generally nonzero. An internal hinge would require a different model.

2 Derive the element stiffness and load vectors

For an element running from node i to node j, define the local coordinate from left to right. The element displacement vector is $[v_i, \theta_i, v_j, \theta_j]^T$. Because both elements have the same global orientation, their local and global bending axes coincide.

$$\mathbf{k}_e = \frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

This matrix follows from the strain energy of an Euler-Bernoulli beam. Cubic Hermite interpolation relates displacement within the element to the four nodal variables. Differentiating twice gives curvature; integrating curvature energy gives the matrix.

$$\mathbf{k}_e = \int_0^L E\,I\,\mathbf{B}^T\mathbf{B}\,dx, \qquad \mathbf{B} = \frac{d^2\mathbf{N}}{dx^2}$$

Consistent loading

A span load contributes equivalent nodal forces and equivalent nodal moments. These are chosen to preserve virtual work for the interpolation functions. The equivalent nodal vector is not the same as the fixed-end member action vector: the latter has the opposite sign.

$$\mathbf{f}_e = \int_0^L \mathbf{N}^T\,q\,dx + \mathbf{N}(a)^T P, \qquad \mathbf{s}_e = \mathbf{k}_e\mathbf{d}_e - \mathbf{f}_e$$

$$\mathbf{f}_{e,q} = q\begin{bmatrix} L/2 \\ L^2/12 \\ L/2 \\ -L^2/12 \end{bmatrix}$$

Both concentrated-force entries P are zero in this exercise, so only the uniform-load term contributes. Support settlement does not enter this load vector: it enters the prescribed displacement vector and the reduced global equation.

3 Evaluate the span loads and assemble the arrays

Insert the actual length and the signed physical load into each element formula. In every load vector below, entries 1 and 3 are in kN, while entries 2 and 4 are in kN m.

$$\mathbf{f}_1 = -30\begin{bmatrix} 11/2 \\ 11^2/12 \\ 11/2 \\ -11^2/12 \end{bmatrix} = \begin{bmatrix} -165 \\ -302.5 \\ -165 \\ 302.5 \end{bmatrix}$$

$$\mathbf{f}_2 = -44\begin{bmatrix} 7/2 \\ 7^2/12 \\ 7/2 \\ -7^2/12 \end{bmatrix} = \begin{bmatrix} -154 \\ -179.666667 \\ -154 \\ 179.666667 \end{bmatrix}$$

What the loop does

```
K = zeros(nDOF);
F = zeros(nDOF,1);
elementK = zeros(4,4,nElement);
elementLoad = zeros(4,nElement);
L = zeros(nElement,1);
% For each element: determine L, form ke and fe, then assemble.
elementK(:,:,e) = ke;
elementLoad(:,e) = fe;
dof = elementDOF(e,:);
K(dof,dof) = K(dof,dof) + ke;
F(dof) = F(dof) + fe;
```

The storage and assembly statements after the comment belong inside the element loop. ke and fe are formed earlier in that loop from the formulas on the previous page. The stored matrices are retained for recovering member end actions after the displacement solution.

K(dof,dof) selects the four global rows and four global columns associated with this element. The plus sign adds contributions rather than overwriting them. At node 2, stiffness and equivalent loads from both spans enter the same global entries. elementK(:,:,e) uses a third array dimension to store one 4 by 4 matrix per element.

In Exercise 15-1 the point-load branch first calculates ξ = a/L, then N, and adds P*N to fe. In Exercise A that branch is skipped because P(e) equals zero. All load signs remain tied to the upward-positive convention.

4 Read the global system and solve its free row

The assembled matrix below follows the degree-of-freedom order [v_1, θ_1, v_2, θ_2, v_3, θ_3]. It is the actual K obtained using EI = 3927 kN m².

Row / column	1	2	3	4	5	6
1	35.4050	194.7273	-35.4050	194.7273	0.0000	0.0000
2	194.7273	1428.0000	-194.7273	714.0000	0.0000	0.0000
3	-35.4050	-194.7273	172.7927	286.1299	-137.3878	480.8571
4	194.7273	714.0000	286.1299	3672.0000	-480.8571	1122.0000
5	0.0000	0.0000	-137.3878	-480.8571	137.3878	-480.8571
6	0.0000	0.0000	480.8571	1122.0000	-480.8571	2244.0000

The table is rounded for reading; the code retains full precision. Matrix entries have mixed units because translation and rotation occupy different columns and force and moment occupy different rows. Symmetry is a useful assembly check.

$$\mathbf{Kd} = \mathbf{F} + \mathbf{R}, \qquad \mathbf{F} = \begin{bmatrix} -165 \\ -302.5 \\ -319 \\ 122.833333 \\ -154 \\ 179.666667 \end{bmatrix}$$

Separate free entries f from prescribed entries c. Only row 4 is needed to find the free rotation. Rows at prescribed degrees of freedom are retained for subsequent reaction recovery.

$$\mathbf{K}_{ff}\mathbf{d}_f = \mathbf{F}_f - \mathbf{K}_{fc}\mathbf{d}_c$$

$$3672\theta_2 + 286.129870(-0.030) = 122.833333$$

$$\theta_2 = \frac{122.833333 + 8.583896}{3672} = 0.03578900584 \text{ rad}$$

The effective right-hand side is 131.417229 kN m. The positive correction comes from subtracting the stiffness coupling to a negative prescribed displacement. Using only F(4) would omit settlement. The 30 mm must first be converted to metres.

```
Kff = K(freeDOF,freeDOF);
Kfc = K(freeDOF,prescribedDOF);
effectiveLoad = F(freeDOF) - Kfc*d(prescribedDOF);
d(freeDOF) = Kff \ effectiveLoad;
```

Backslash solves the reduced linear system. Explicitly forming inv(Kff) is unnecessary. The original unconstrained K is singular because rigid motions have not yet been removed; the reduced system contains the support constraints.

5 Reproduce the solution analytically

Use slope-deflection to obtain the same end moments without assembling a global matrix. Define M_{ij} as the action applied to the member at its i end, with counterclockwise positive at both ends. Define the chord displacement $\Delta = v_j - v_i$, measured upward. This convention matches endAction in the code.

$$M_{ij} = \frac{2EI}{L}\left(2\theta_i + \theta_j - \frac{3\Delta}{L}\right) + M_{ij}^F$$

$$M_{ji} = \frac{2EI}{L}\left(\theta_i + 2\theta_j - \frac{3\Delta}{L}\right) + M_{ji}^F$$

The fixed-end terms M^f are the actions required when both ends of an element are held at zero displacement and rotation. They equal minus the rotational entries of fe. Set $\theta_1 = \theta_3 = 0$, and enforce $M_{21} + M_{23} = 0$ because there is no externally applied couple at node 2.

For span 1, $\Delta_1 = v_2 - v_1 = -0.030$ m. For span 2, $\Delta_2 = v_3 - v_2 = +0.030$ m. The two chord movements therefore have opposite signs. The downward UDL fixed-end actions are [+302.5, −302.5] and [+179.666667, −179.666667] kN m.

$$M_{12} = 714\theta_2 + 308.341818, \qquad M_{21} = 1428\theta_2 - 296.658182$$

$$M_{23} = 2244\theta_2 + 165.240952, \qquad M_{32} = 1122\theta_2 - 194.092381$$

$$M_{21} + M_{23} = 3672\theta_2 - 131.417229 = 0$$

$$\theta_2 = 0.03578900584, \quad M_{12} = 333.895168, \quad M_{32} = -153.937116$$

All end moments above are in kN m. This hand calculation has the same scalar free-rotation equation as the matrix solution. The agreement checks the fixed-end loads, support settlement signs, and joint equilibrium without relying on the assembly loop.

6 Recover member end actions and support reactions

Insert the solved displacement vector into every element. Recover the element actions by subtracting the equivalent span-load vector. Then assemble the support reactions from Kd − F. A displacement boundary condition creates an unknown reaction; an unconstrained rotation cannot create a support reaction moment.

$$\mathbf{s}_e = \mathbf{k}_e\mathbf{d}_e - \mathbf{f}_e, \qquad \mathbf{R} = \mathbf{K}\mathbf{d} - \mathbf{F}$$

```
R = K*d - F;
for e = 1:nElement
    de = d(elementDOF(e,:));
    endAction(:,e) = elementK(:,:,e)*de - elementLoad(:,e);
end
M_support = [-endAction(2,1); endAction(4,1); endAction(4,2)];
```

Each endAction column contains $[V_i, M_i, V_j, M_j]^T$. The local nodal displacements de are selected from d using elementDOF. The two end forces at node 2 add to its vertical reaction; the two end moments at that node cancel.

The vertical member actions can also be recovered by hand from moment and vertical-force equilibrium of each span. With signed q and P, use the following expressions before adding the member forces at the roller.

$$V_j = -\frac{M_i + M_j + qL^2/2 + Pa}{L}, \qquad V_i = -qL - P - V_j$$

Element	V_i in kN	M_i in kN m	V_j in kN	M_j in kN m
1	173.031244	333.895168	156.968756	-245.551481
2	167.087766	245.551481	140.912234	-153.937116

Support	Vertical reaction in kN	Reaction moment in kN m
1	173.031244	333.895168
2	324.056522	0.000000
3	140.912234	-153.937116

Convert external end actions to the internal diagram

$$M(0) = -M_i, \qquad M(L) = M_j$$

Support	Internal bending moment in kN m
1	-333.895168
2	-245.551481
3	-153.937116

The left support reaction moment and the left internal diagram moment have opposite signs because they act on different free bodies. All three support-section internal moments here are negative, meaning hogging. The right support reaction is clockwise because R(6) is negative.

7 Derive shear force and bending moment between nodes

For a cut at local position x, use equilibrium of the segment to the left of the cut. A point force contributes only after its application point. The step H is zero before the force and one after it; the ramp bracket means max(x − a, 0).

$$V(x) = V_i + qx + PH(x-a), \qquad M(x) = -M_i + V_i x + \frac{qx^2}{2} + P\langle x-a\rangle$$

Let x run from 0 to 11 m along span 1. In span 2 use z = global x − 11 m, so z runs from 0 to 7 m. Shear is linear and moment is quadratic within both uniformly loaded spans.

$$V_1(x) = 173.031244 - 30x, \quad M_1(x) = -333.895168 + 173.031244x - 15x^2$$

$$V_2(z) = 167.087766 - 44z, \quad M_2(z) = -245.551481 + 167.087766z - 22z^2$$

At the roller, shear rises from −156.968756 to +167.087766 kN. The jump is +324.056522 kN, matching its vertical reaction. Moment stays continuous at −245.551481 kN m.

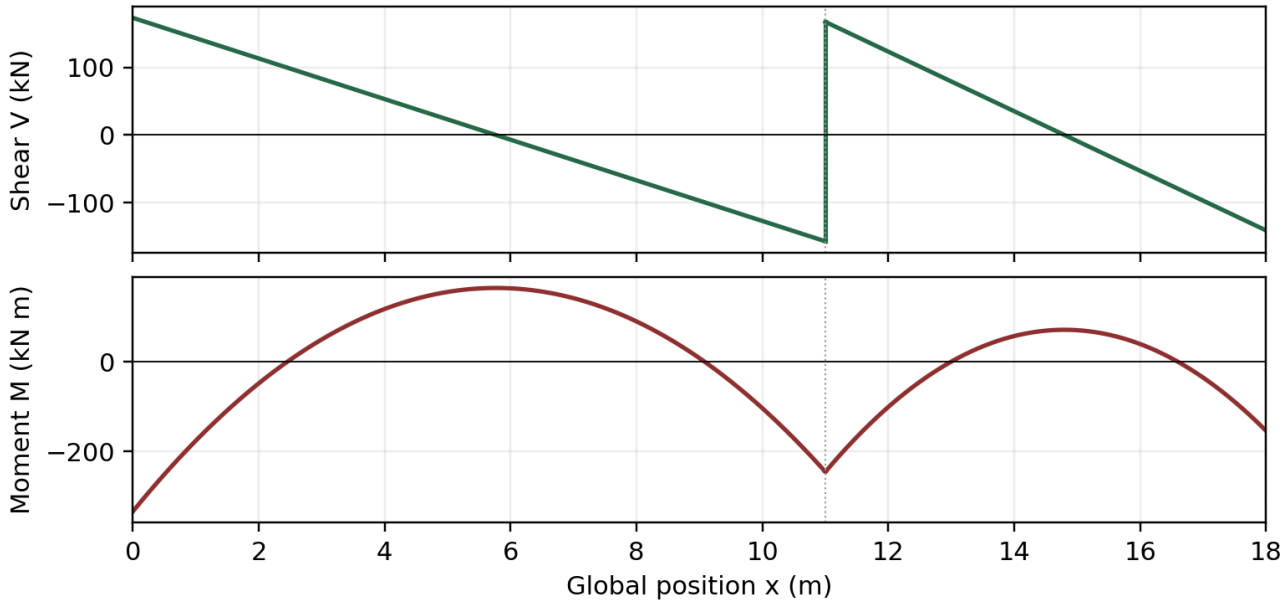


Figure 2. Response calculated from span equilibrium. Moment is positive in sagging; the negative support values indicate hogging.

8 Integrate bending moment to recover deflection

With the chosen signs, EI times the second derivative of displacement equals internal bending moment. Integrate once for rotation and twice for displacement. The known left-end rotation and displacement supply the integration constants.

$$EI\, v''(x) = M(x)$$

$$\theta(x) = \theta_i + \frac{-M_i x + V_i x^2/2 + q x^3/6 + P\langle x-a\rangle^2/2}{EI}$$

$$v(x) = v_i + \theta_i x + \frac{-M_i x^2/2 + V_i x^3/6 + q x^4/24 + P\langle x-a\rangle^3/6}{EI}$$

The MATLAB arrays x, ramp, V, M, and v store values throughout each element. A dot before a power, as in x.^2, means element-by-element exponentiation. This is required because x is a vector. Cell arrays xPlot{e} and vPlot{e} store vectors that can have different lengths for different elements.

```
ramp = max(x-a(e),0);
M = -s(2) + s(1)*x + q(e)*x.^2/2 + P(e)*ramp;
v = de(1) + de(2)*x + ...
    (-s(2)*x.^2/2 + s(1)*x.^3/6 + q(e)*x.^4/24 + ...
     P(e)*ramp.^3/6)/EI;
```

A curve obtained only by joining nodal displacements would miss bending under the span loads. The integration above retains those loads. It is exact for this ideal prismatic beam model, rather than an approximation limited to a cubic curve between solved nodal values.

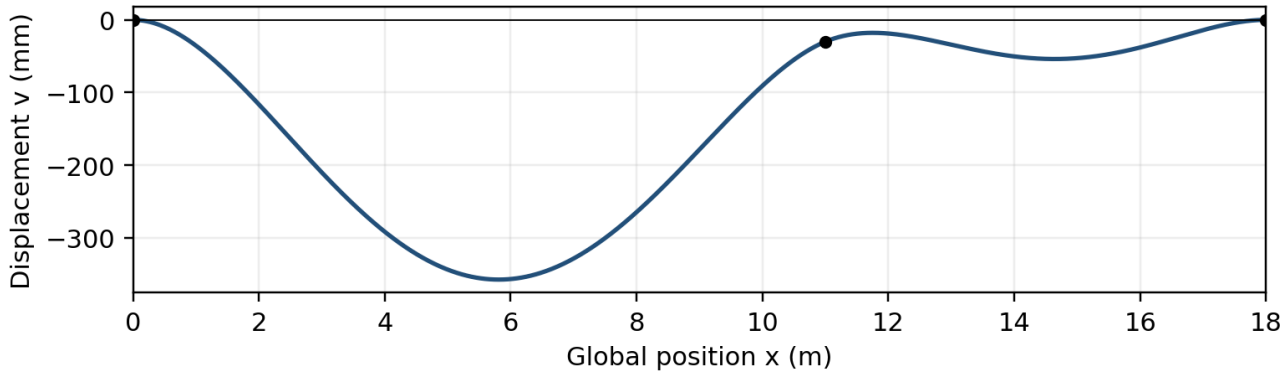


Figure 3. Physical displacement in millimetres. Support 2 is at −30 mm; fixed-end displacement and rotation are zero.

At each element right end, the recovered displacement and rotation must match de(3) and de(4). The code checks this compatibility. It provides a stronger check than simply plotting a smooth-looking curve.

9 Separate loading and support settlement

Because the beam model is linear, solve two simpler problems and add their results. In the first, retain the physical loads but prescribe zero support displacement. In the second, set the physical loads to zero and impose only the 30 mm settlement.

$$\mathbf{d} = \mathbf{d}_{\text{load}} + \mathbf{d}_{\text{settlement}}, \qquad \mathbf{R} = \mathbf{R}_{\text{load}} + \mathbf{R}_{\text{settlement}}$$

```
d_load = zeros(nDOF,1);
d_load(freeDOF) = Kff \ F(freeDOF);
R_load = K*d_load - F;
d_settlement = zeros(nDOF,1);
d_settlement(prescribedDOF) = d(prescribedDOF);
d_settlement(freeDOF) = Kff \ (-Kfc*d_settlement(prescribedDOF));
R_settlement = K*d_settlement;
```

No F is subtracted in the settlement-only reaction expression because there are no applied loads in that subproblem. The settlement-induced member forces still arise through the nonzero displacement field.

Reaction	Loads only	Settlement only	Combined
V_1 in kN	171.513889	1.517355	173.031244
M_1 in kN m	326.384259	7.510909	333.895168
V_2 in kN	328.571429	-4.514906	324.056522
V_3 in kN	137.914683	2.997551	140.912234
M_3 in kN m	-142.134259	-11.802857	-153.937116

Why EI matters differently in the two parts

For loads alone, multiplying a common EI by a factor changes all solved rotations and deflections by its reciprocal, leaving reactions and internal moments unchanged. For settlement alone, the imposed displacement is held fixed: multiplying EI increases the displacement-induced actions by that factor.

The combined answer therefore depends on the specified EI. Replacing it by the reference value 1, as done in Exercise 15-1 with no support movement, would give a different and incorrect settlement contribution here.

The script checks displacement and reaction superposition with assert. Reversing the settlement sign reverses the settlement-only contribution, provided the same linear support model remains applicable.

10 Verify equilibrium and compatibility

A numerical result should satisfy the original physical problem, not just the matrix equations used to generate it. Check whole-beam equilibrium using the actual forces and load resultants. Do not add equivalent nodal loads to those same physical loads, because that counts the loading twice.

$$\sum F_y = 173.031244 + 324.056522 + 140.912234 - 330 - 308 = 0$$

$$\sum M_1 = 333.895168 - 153.937116 + 324.056522(11) + 140.912234(18) - 330(5.5) - 308(14.5) = 0$$

$$M_{21} + M_{23} = -245.551481 + 245.551481 = 0$$

The displayed numbers are rounded, so hand substitution may leave a small rounding residual. The script evaluates these checks with full precision and uses numerical tolerances rather than requiring exact floating-point zero.

Checks implemented in the script

Check	What it tests
K equals its transpose	Reciprocity and correct symmetric assembly
Vertical and moment residuals are near zero	Whole-beam force and moment equilibrium
R(freeDOF) is near zero	No reaction couple at an unrestrained rotation
Two member moments at node 2 add to zero	Joint moment equilibrium
Recovered v and theta meet right-end nodal values	Displacement and rotation compatibility
M_support matches referenceMoments	Original exercise agrees with the hand solution

```
assert(norm(R(freeDOF),inf) < 1e-8, ...
    'An unrestrained rotation cannot develop a support reaction moment.');
assert(abs(endAction(4,1)+endAction(2,2)) < 1e-8, ...
    'Member end moments must balance at the unloaded roller joint.');
```

When intentionally changing geometry, loads, or settlement, disable the comparison with the original referenceMoments. Keep the equilibrium and compatibility checks. In Exercise B, changing EI also changes the combined reference answer.